

# Smaller and Shallower XOR Circuits for AES MixColumns: 97 Gates at Depth 3; 92 at Depth 4; 88 at Depth 5

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## Abstract

We give eight verified implementations of AES MixColumns as circuits of 2-input XOR gates over  $\text{GF}(2)$ . Four improve the published depth–count Pareto frontier at their depth, on lineages holding no imported circuit: **97** gates at depth **3**, the minimum depth, against 99 [2]; **92** at depth **4** against 97 [3]; **88** at depth **5** against 94 [3]; and **88** at depth **6** against 92 [4, 5]. **88 is not a new count**: it is Jean’s published count [9], and **Jean has priority**. What the depth-5 and depth-6 circuits improve is the depth at that count. Version 3 of this note reported two frontiers, because its only 88 at depth 5 had a seed chain through Jean’s circuit; a circuit found from scratch now reaches that point, so the frontier is one line, **97 @ 3, 92 @ 4, 88 @ 5**, entirely on our own lineage. *That removes our dependence on Jean’s circuit at the depth-5 point; it does not beat it.* Also here: an 88 at depth 7 that *ties* Jean’s point with an independent circuit sharing 61 of 88 internal masks, an 89 at depth 5 now dominated, and two further 88s at depths 5 and 8 derived from Jean’s circuit and reported as such. All were found by stochastic local search over a value-set formulation of the shortest-linear-program problem. No 87 was found: 47 canonical 88-gate circuits, and both from-scratch 88s here, have exhaustively empty remove- $\leq 3$  neighbourhoods. Nothing here is claimed optimal.

## 1 Model and specification

A circuit is an ordered list of 2-input XOR gates:  $s_0, \dots, s_{31}$  are the input bits, and gate  $k$  produces  $s_{32+k} = s_a \oplus s_b$  with  $a, b < 32 + k$ . Cost is the gate count; depth counts gates on the longest input-to-output path, inputs at depth 0. Bit index  $i$  denotes bit  $(i \bmod 8)$  of state byte  $\lfloor i/8 \rfloor$ , least-significant-bit first, and MixColumns maps an input column  $a_0, \dots, a_3$  to

$$\text{out}[c] = 2 \cdot a_c \oplus 3 \cdot a_{(c+1) \bmod 4} \oplus a_{(c+2) \bmod 4} \oplus a_{(c+3) \bmod 4},$$

in  $\text{GF}(2^8)$  modulo  $x^8 + x^4 + x^3 + x + 1$  (0x11b), per NIST FIPS 197-upd1, Eq. 5.6 [1]. The map is  $\text{GF}(2)$ -linear, so agreement on the 32 unit-input vectors is a complete correctness check; counts and depths are relabeling-invariant, so no comparison below depends on convention.

The s-XOR (in-place) model is comparable: SSA-unrolling a  $k$ -instruction s-XOR program yields exactly  $k$  free-standing gates. Multi-input XOR gates, gate-equivalent area and quantum CNOT counts are different cost functions: not comparable, not claimed against.

## 2 Results

Depth 3 is the known minimum depth — an output depending on  $w$  inputs needs depth  $\geq \lceil \log_2 w \rceil$ , and MixColumns has outputs of weight 7 [2] — so the depth-3 circuit’s contribution

| Depth          | Published best | Model        | This work         | Relation                                 |
|----------------|----------------|--------------|-------------------|------------------------------------------|
| 3              | 99 [2]         | XOR2         | <b>97</b>         | −2 (depth 3 is the minimum depth)        |
| 4              | 97 [3]         | XOR2         | <b>92</b>         | −5                                       |
| 5              | 94 [3]         | XOR2         | <b>88</b>         | −6, from scratch; dominates [9]          |
| 5              | 94 [3]         | XOR2         | 89                | −5; dominated here                       |
| 5              | 94 [3]         | XOR2         | (88) <sup>†</sup> | −6, derived from [9]                     |
| 6              | 92 [4, 5]      | XOR2 / s-XOR | 88                | −4, from scratch; dominated here         |
| 7 <sup>‡</sup> | 88 [9]         | XOR2         | 88                | ties it; independent, 61/88 masks shared |
| 8              | —              | XOR2         | (88) <sup>†</sup> | derived from [9]; dominated              |
| 9 <sup>‡</sup> | 89 [8]         | XOR2         | —                 | off the frontier                         |

Table 1: The published depth–count landscape in comparable models, and this work. Our frontier is one line, 97 @ 3, 92 @ 4, 88 @ 5, and no imported material lies anywhere in its lineage. 88 is *not* a new count: it is Jean’s [9], who has priority; the depth-5 and depth-6 rows improve the depth at that count, not the count. <sup>†</sup>Derived: the seed chain passes through [9], which is credited; not an independent construction. <sup>‡</sup>*Neither* [9] *nor* [8] states a depth: the 7 and the 9 are both our own measurements of our own transcriptions, on the same footing, and both are *forced* — the ASAP least-fixpoint schedule over each published mask set, the shallowest either admits, still gives 7 and 9. [8] reports its count as “g-XOR”, which is the same model as the XOR2 of every other row (Section 1). Jean’s 88 [9] supersedes the earlier 91 of [6, 7] at depth 7; the depth-3 lineage is 105 → 103 → 102 → 99 ([2], Table 3). Listings: Appendices A–H.

is its count. The 89 at depth 5, five gates below the published depth-5 point, is dominated at its own depth within this note and kept for the record.

The 88 at depth 7 *ties* Jean’s point rather than improving it: Jean’s note was posted 2026-07-23 and has priority. Ours was found 2026-07-26 on a fully logged own lineage — from-scratch 97 at depth 3 (235s), a 21 h ladder to 89 at depth 6 (11.0h), one reroute to 89 at depth 5 (592s),  $\rho^2$ -symmetrization to a 94 at depth 5, then 32.9min of plateau walking — with no imported circuit in the chain, and is a different circuit: 61 of 88 masks shared, Jaccard 0.530. For scale, Jean’s 88 and the 89 of [8] — two independently published circuits — share 63; overlap of that size is what independence looks like for this map. The 88 at depth 8 (Appendix H) is derived work and dominated: its seed chain passes through Jean’s,  $\rho^2$ -symmetrized and peeled to 95, orbit-walked to 92, then unioned with a 91 of our own lineage before the descent.

Three further 88s differ in exactly the way that matters, and two are **from scratch**: the **88 at depth 6** (Appendix F), found 2026-07-28 by worker `c_naive` at restart 18 of its fourth session, and the **88 at depth 5** (Appendix C), found 2026-07-30 by the same worker at restart 16 of its fifth, whose root `constructors.build("naive", 1958)` — a randomized balanced XOR tree over the 32 raw inputs — rebuilds to exactly the logged 146-gate depth-3 circuit; from there the walk reached 88 gates at depth 6 at iteration 33 873 and 88 at depth 5 at iteration 37 155, six seconds later on the Pareto depth tie-break.

The reason nothing imported can have reached either is the *root*, not a knob setting: the constructor branch of the archived worker’s root generator calls only `constructors.build` and has no file-reading path at all, while every branch that can open a circuit off disk is a different branch, unreachable under this root spec. The producing engine reads nothing either: both finds came from walk chunks, and the walk has no candidate pool and no disk read path. Knobs close the rest — the cross-pollination flag has one writer in the whole repository, the hunt worker never sets it, the supervisor never launches the process that would, and the untouched logs carry none of the lines a merge emits; repulsion was off; each restart carries its own context, visibly, since restart 16 opens worse than restart 15 ended. Knobs alone

would not settle it, as the derived 88 below shows: identical knobs, and imported material still reached it through its root.

The depth-6 circuit was the first 88 we found from scratch, and undercuts the published depth-6 92 [4, 5] by four gates. The depth-5 one supersedes it and does something else besides: until it was found, the (88, 5) point here was held only by a circuit whose seed chain runs through Jean’s published work, which is why version 3 reported two frontiers. **This removes our dependence on that circuit at the depth-5 point; it does not beat it**, and 88 remains Jean’s count, with priority. What it does improve is published: six gates below the depth-5 94 [3], and it dominates Jean’s 88 — equal count, two levels shallower, on the *forced* depth 7 of Table 1, note ‡. Its own depth 5 is forced in the same sense, 11 output bits sitting there. It dominates our 88s at depths 6, 7 and 8 and our 89 at depth 5. “No imported material” claims less than it may seem: not method independence, since the engines were tuned over a campaign that did read published circuits; not disjointness, since 42 of its 88 masks are Jean’s too, though 32 of those are the forced targets; and not a reproducible descent — the root is exact, the trajectory is not. It sits outside every family named here: its Jaccard to every other circuit in this repository is 0.285–0.323, far below the 0.7 same-family threshold, and it is no closer to any of them than to its own rotations; its largest similarity to anything measured is 0.386, at  $\rho^3$  of Jean’s 88. Nor is it a lone point: 135 distinct own-lineage 88-gate depth-5 mask sets came out of the same pocket, all oracle-verified.

The **88 at depth 5** of Appendix E, found 2026-07-29, is **derived work**, and that is the first thing to say about it: its seed chain passes through Jean’s 88 — symmetrized and peeled to 95, orbit-walked to 92, unioned with a 91 of our own lineage, descended to the exactly  $\rho^2$ -symmetric 90 at depth 9 of Section 4, reseeded mask-identically into the orbit hunt, then polished to 88 at depth 6 and 88 at depth 5, with mask identity checked at every link. It is six gates below the published depth-5 94 [3]. Neither is it a new construction: 75 of its 88 masks are the depth-5 89’s (Jaccard 0.735, the largest overlap between any two circuits here), so it is that 89’s own basin reached at 88 gates, not a new family. It is retained, superseded at its point rather than withdrawn.

One frontier therefore: **97 @ 3, 92 @ 4, 88 @ 5**, every point of it on our own lineage with no imported material. **87 was not found**.

### 3 Method

A circuit is represented as a *value set*: masks over the 32 inputs, valid iff every mask is the XOR of two already-available masks and all 32 output masks are present; gate count is the set size. Deletion becomes a set operation, circuits compare by mask overlap regardless of gate order, and a neighbourhood — remove  $k$  masks, add at most  $k - 1$  — becomes finite and enumerable (Section 4).

Good circuits at this size *look* locally efficient — every gate reads as an optimal step — and that is what makes them hard to improve: they sit at local optima, so smaller circuits are separated from them by locally *suboptimal* steps. (This was true of the naive construction we started from and of every circuit we produced up to the 89; we make no claim about how anyone else’s circuit was built.) **Remove-1** drops a non-output mask if the set stays realizable. The **neutral swap** removes one mask and adds one repair mask at unchanged size: with  $A = \text{closure}(S')$  and  $\text{stuck} = (S' \cup T) \setminus A$ , every single-mask repair lies in  $C = \{u \oplus a : u \in \text{stuck}, a \in A\} \cap P_2(A)$  ( $P_2$  = pair sums), so enumerating  $C$  *decides* the move — measured  $|C| \approx 7$ , cheaper than the earlier engine’s 24–40 random candidates, which missed 81–82 % of existing repairs. **Remove-2-add-1** remains a diversifier, though at the 88/89 frontier it is provably empty (Section 4). **Destroy-and-rebuild** removes a group and re-synthesizes from a pool, accepting rebuilds no larger, or larger on an annealing schedule with reheat; destroying

a *connected cone* of the DAG leaves a hole worth re-planning, where  $k$  unrelated masks are usually just re-derived, and near-miss rebuilds are peeled before judgement. A fixpoint over minimum build depths rejects cap violations and emits gate lists in depth order.

Workers run one engine at one depth cap in fixed-length chunks, alternating walk and rebuild, and a coordinator re-offers each the best circuit feasible at its cap (the reseeding ladder). A worker accepts fewer gates *or equal gates at strictly lower depth* — a Pareto tie-break, and not a cosmetic one: 89 at depth 6 became 89 at depth 5 through it; both 88s of version 2, first seen at depths 11 and 10, reached depths 7 and 8 seconds later; the 88 at depth 6 followed its own depth-7 predecessor half a second later; and the from-scratch 88 at depth 5 followed its own depth-6 predecessor six seconds later. Every candidate is verified against MixColumns rebuilt from  $\text{GF}(2^8)$  before acceptance. The depth-3 circuit comes from a separate constructor: each output is modelled as  $A \oplus B$ ,  $A$  and  $B$  partitioning its bits into shared, recount-costed depth- $\leq 2$  parts, annealed by iterated local search, so depth  $\leq 3$  holds by construction.

**The v2 engine.** Version 1’s circuits came from an engine whose kernel was then profiled and rewritten [10]: level-BFS depth relaxation ( $17.5\times$  on destroy-and-rebuild, bit-identical trajectories); a worklist closure with incremental removal queries, and removal of a bug that re-verified unchanged circuits every plateau iteration — 78% of walk runtime ( $21.9\times$  on the walk); victim repooling ( $\sim 34\times$  accepted moves/s); cone destroys ( $\sim 12\times$ ); exact repair ( $\sim 7\times$  plateau mobility). The walk now runs at 480–640 iterations/s against 70–110. What found both 88s was not a cleverer move but a cheap plateau walked at volume: every distinct equal-best mask set is harvested.

## 4 Exact local certificates, and structure

To say what is *not* there we decide the neighbourhood question: given a verified 88-gate circuit and a window of  $k$  removed masks, can at most  $k - 1$  new masks — each buildable when added, with cascade unlocks between — restore all 32 targets? A “yes” is an 87-gate circuit; a completed “no” is a proof for that window. Completeness follows from a first-unlock case analysis: the first unlock must have the newest added mask as a parent, which pins each added mask to a finite, enumerable shape. Runs above the proved budget are recorded as undecided, never as irreducible. The procedure agrees with an independent brute force 25/25 and 12/12 at  $k \leq 3$ , and on 1 257 instances including 122 genuine negatives at  $k = 4$ .

- **47 canonical 88-gate circuits** — Jean’s 88, its 12 plateau siblings, and a third family’s anchor with 33 representatives — **have exhaustively empty  $k \leq 3$  shells**: all 1 540 windows at  $k = 2$  and all 27 720 at  $k = 3$  each, none reducible. Hence *any 87-gate circuit differs from each of the 47 by at least 4 masks*. Stated plainly: all 47 lie in Jean’s-lineage families — families 1–2 are Jean’s circuit and its siblings, and the third family’s anchor is itself seeded through Jean’s circuit — so this is a rigidity statement about *that* neighbourhood, and our 88 at depth 7, independent of that lineage, is exactly the least-certified: it is *not* among the 47, its exhaustive  $k \leq 3$  sweep was never run, and it has only 9 exact  $k = 4$  windows and 8 SAT cone windows. Version 2 of this note recorded that the best-certified shells and the independent one were therefore disjoint; that no longer holds. **Both from-scratch 88s — at depths 6 and 5, outside Jean’s lineage — have exhaustively empty  $k \leq 3$  shells of their own** (all 1 540 windows at  $k = 2$  and all 27 720 at  $k = 3$ , each), so any 87 differs from each of them too by at least 4 masks. The *derived* depth-5 88’s  $k = 2$  shell is empty as well (1 540 windows, so any 87 differs from it by at least 3 masks), but its  $k = 3$  shell was *never swept*: with the depth-7 88 it is one of the two least-certified circuits here.

- **Population.** Harvesting mapped  $\approx 139\,878$  distinct 88-gate mask sets, of which 105 801 are proven irreducible at  $k = 2$ . In total  $\approx 165$  million exact window decisions, zero timeouts, zero reducible windows; all five 89-gate seeds likewise have empty  $k = 2$  shells.
- **A retracted observation, and the certificate that survives it.** Version 2 of this note observed that no 88-gate mask set of minimum build depth  $\leq 6$  had been seen in the harvesting. The two circuits added here, at depths 6 and 5, refute that observation — which was reported as an observation and not as a certified claim. What was certified is narrower and stands: near the *old* 88-plateau no 88 at depth 6 exists, all 4 861 two-critical cores being UNSAT at cap 6. The new circuits live in different basins, and the basins differ in exactly the relevant place: the old plateau’s depth obstruction is output rows 3 and 27, the depth-6 88’s is rows 1, 11, 17 and 25 — four weight-7 targets, three of them column-rotations of one another — and the from-scratch depth-5 88’s is a third pattern again, rows 1, 7, 12, 13, 17, 18, 21, 25, 27, 28 and 31. MixColumns has at least three structurally distinct ways to hit a depth wall at 88 gates, and they give way at different depths.
- **A negative result that predicted where to look, and was right.** Whether the depth-6 88’s own family contains a member realizable at depth 5 was answered first, and negatively. All 8 993 members of that family were screened by their ASAP (least-fixpoint) depth — the shallowest depth each mask set admits at all — and the histogram is exactly 4 420 at depth 6 and 4 573 at depth 7, with *nothing below 6*: depth 6 is minimal for every member, not merely attained. A depth-capped-5 SAT sweep of 27 windows around the circuit returned 22 UNSAT and **0 SAT**, the 5 undecided ones all at the widest radius. We recorded the conclusion as “a Jean-independent 88 at depth 5, if one exists, lives somewhere else”. It does, and it does: the from-scratch 88 at depth 5 is in neither basin, sharing 43 of 88 masks with the depth-6 88 and 42 with the derived depth-5 one.
- **Windowed SAT is evidence, not proof.** A sound windowed CNF encoding, solved by Kissat under two soundness-proved symmetry-breaking layers, returned UNSAT up to  $k = 16$  on Jean’s 88 and  $k = 15$  on the third-family anchor (whose whole  $k \leq 12$  regime is UNSAT, 28/28), with **0 SAT anywhere** — but UNSAT there is relative to the encoding’s fixed slot order, and undecided windows are undecided.

None of this bounds 87 away globally; it is consistent with 88 being locally rigid, and it redirected the search: at 88 gates every move class small enough to enumerate is empty, so what remains is walking the equal-size plateau widely.

**Symmetry.** Byte rotation  $\rho$  (order 4) commutes with MixColumns, so the 32 targets form 8  $\rho$ -orbits of size 4 and equivariant circuits can be searched in orbit space — but full  $\rho$ -symmetry costs about +19 gates. The half-rotation  $\rho^2$  is where the structure is: in the elite basin 79–81 % of a circuit’s masks have their  $\rho^2$ -image also in the mask set, and the 12  $\rho^2$ -fixed masks the search settles on are exactly the classic  $x_0 \oplus x_2 / x_1 \oplus x_3$  pairs, rediscovered rather than imposed. Jean’s 88, at 75 % (66 of 88), symmetrizes into a second, distinct basin. A  $\rho^2$ -equivariant orbit engine produced two *exactly*  $\rho^2$ -symmetric 90-gate circuits (depths 9 and 7), improving the best exactly symmetric circuit we know of from 94 to 90; both are certified locally optimal in orbit space, and both are derived work, each seed union containing a circuit symmetrized from Jean’s 88; the depth-9 one is the ancestor of the derived 88 at depth 5 (Section 2). Symmetry is also a seeding device: the exactly  $\rho^2$ -symmetric 94 at depth 5 — 82 of its 94 masks shared with the 89 at depth 5 — costs +5 over that 89 and is the seed from which the 88 at depth 7 was found.

## 5 Verification, artifacts, and reproduction

Every circuit here is checkable from the note alone: the appendices list every gate, Section 1 defines the target map, and because MixColumns is linear, replaying a listing on the 32 unit-input vectors against Eq. 5.6 [1] is a complete correctness check. Each appendix also states its circuit’s canonical SHA-256 over the UTF-8 bytes of `{"inputCount":32,"gates":[...]}`.

The artifacts automate that. Circuits and verifiers are at <https://github.com/Joe-b-20/aes-mixcolumns-xor-circuits>, archived at concept DOI 10.5281/zenodo.21299092; `python3 verify_all.py` runs `verify.py`, which rebuilds the target map from the  $\text{GF}(2^8)$  definition so the verifier *is* the specification, a clean-room verifier with adversarial rejection tests, and Verilog testbenches, all under continuous integration. The method, pipeline, untouched run archives with each run’s exact code, certificate summaries and the credited transcriptions of the imported circuits [9, 8] are published separately [10], with a reproduction command for every result that has one — neither from-scratch 88 has one, and the derived 88s at depths 5 and 8 are deliberately offered as archives rather than recipes. Measured there on 2026-07-27 with the shipped v2 engine, and reported as measurements rather than promises: the from-scratch 97 at depth 3 in 81 s on one core; the 89 at depth 5 in 19 s and 22 s from shipped seeds, against 592 s for the archived 2026-07-14 run on the v1 engine; the 88 at depth 7 in 19.4 and 31.0 minutes in two single-worker re-runs from the  $\rho^2$ -symmetric 94, against 32.9 min in the archived ten-worker run; the 92 at depth 4 a from-scratch cascade of hours. The re-runs are re-runs, not independent confirmations.

Implementation and execution used LLM coding agents as programming tools, directed by the author, who designed the moves and acceptance rules. Concretely: the 97 at depth 3, the 92 at depth 4 and the 89 at depth 5 are reproduced by the published, dependency-free code with no AI system in the loop, as is a single-worker re-run of the 88 at depth 7. *All five* 88-gate circuits came out of author-directed LLM-agent campaigns, in two chapters — the v2 engine, the certificates of Section 4 and the 88s at depths 7 and 8 from a 24-agent campaign over 2026-07-26/27, and the three 88s at depths 6 and 5 from a later multi-day fleet of sixteen processes running that engine, built and operated by the same agents, on 2026-07-28, 2026-07-29 and 2026-07-30. The two from-scratch 88s are the frontier records of this note with *no* single-command reproduction: each came out of one restart of one worker, 18.96 h and 19.22 h into its session. What is published for them instead is each root constructor and seed, which rebuild the logged 139-gate and 146-gate depth-3 starting circuits exactly, and the worker’s untouched log of every chunk from there to 88 gates. That distinction changes nothing about correctness: every circuit any of it produced is machine-verified against MixColumns rebuilt from  $\text{GF}(2^8)$ , and the method is fully specified and reproducible from the accompanying code either way.

## 6 Scope and claims

- **Not optimality.** Minimum 2-input-XOR circuit size (the Shortest Linear Program problem) is NP-hard; no count here is proven minimal, and the certificates of Section 4 are local.
- **Comparisons.** All counts are for the model and convention of Section 1; the baselines [2, 3, 4, 5, 6, 7, 8, 9] were source-checked, and a source-by-source audit, with a dated Corrections section, is in the artifact repository’s `PRIOR_ART.md`. We claim the smallest counts at depths 3, 4, 5 and 6, and no improvement at unconstrained depth: the published floor is 88 and stays 88.
- **Priority.** 88 is Jean’s count [9], posted first; none of our five 88s is a count record.

The one at depth 7 ties that point independently; the from-scratch ones at depths 5 and 6 improve the depth at that count, and the depth-5 one removes our own dependence on Jean’s circuit at that point rather than improving on it; the remaining two, at depths 5 and 8, are derived from Jean’s circuit, credited, and reported as derived wherever they appear. The depths quoted for [9] and [8] — 7 and 9 — are ours, not theirs, and are treated identically: see Table 1, note ‡.

- **Earlier results.** This note supersedes the project’s 98 at depth 3, 91 at depth 6 and 89 at depth 10, kept in the artifact repository.

**Version history.** **v3.1**, 30 July 2026: adds a second from-scratch 88, at depth 5, reaching with nothing imported the point v3 reached only through derived work; the two frontiers of v3 collapse into one. 88 remains Jean’s count, with priority; no claim of v3 is withdrawn and both of its depth-5 circuits are retained. **v3**, 29 July 2026: adds the 88 at depth 6 (the first 88 here found from scratch) and the derived 88 at depth 5, both frontier statements, the depth-obstruction datum, and the two new  $k \leq 3/k = 2$  shells. Two *observations* of v2 are retracted, both refuted by these circuits: that no 88-gate mask set of build depth  $\leq 6$  had been seen, and that the best-certified 88-gate shells and this project’s Jean-independent 88 were disjoint sets (the depth-6 88 is now in both). No *claim* of v1 or v2 is withdrawn. **v2**, 27 July 2026: adds the 88 at depth 7 and its derived depth-8 companion, the exact neighbourhood certificates, the  $\rho^2$  symmetry results, the v2 engine measurements and dated reproduction times; no claim of v1 is withdrawn. **v1**, July 2026 (the 97, 92 and 89 only), remains available in the repository history and in the Zenodo version archive under the concept DOI 10.5281/zenodo.21299092.

## References

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## A Circuit mixcolumns\_97gates\_depth3 (97 gates, depth 3)

Signals  $s_0, \dots, s_{31}$  are the inputs; gate  $k$  produces  $s_{32+k}$ . Canonical gate-list SHA-256 (see Section 5):

e4c4e678c4b7c22d889c6e060b43859c111b82b953d6d9b43c49c079c0545b56

|                        |                        |                         |                         |
|------------------------|------------------------|-------------------------|-------------------------|
| s32 = s0 $\oplus$ s8   | s57 = s7 $\oplus$ s31  | s82 = s51 $\oplus$ s41  | s107 = s75 $\oplus$ s18 |
| s33 = s5 $\oplus$ s13  | s58 = s23 $\oplus$ s31 | s83 = s53 $\oplus$ s3   | s108 = s80 $\oplus$ s66 |
| s34 = s6 $\oplus$ s14  | s59 = s33 $\oplus$ s14 | s84 = s52 $\oplus$ s40  | s109 = s84 $\oplus$ s66 |
| s35 = s7 $\oplus$ s15  | s60 = s35 $\oplus$ s17 | s85 = s52 $\oplus$ s21  | s110 = s62 $\oplus$ s55 |
| s36 = s8 $\oplus$ s16  | s61 = s39 $\oplus$ s10 | s86 = s42 $\oplus$ s54  | s111 = s86 $\oplus$ s89 |
| s37 = s9 $\oplus$ s17  | s62 = s41 $\oplus$ s5  | s87 = s53 $\oplus$ s54  | s112 = s93 $\oplus$ s64 |
| s38 = s10 $\oplus$ s18 | s63 = s35 $\oplus$ s40 | s88 = s43 $\oplus$ s56  | s113 = s67 $\oplus$ s58 |
| s39 = s11 $\oplus$ s19 | s64 = s42 $\oplus$ s7  | s89 = s55 $\oplus$ s30  | s114 = s94 $\oplus$ s72 |
| s40 = s11 $\oplus$ s20 | s65 = s36 $\oplus$ s43 | s90 = s57 $\oplus$ s0   | s115 = s75 $\oplus$ s71 |
| s41 = s12 $\oplus$ s20 | s66 = s43 $\oplus$ s19 | s91 = s57 $\oplus$ s3   | s116 = s95 $\oplus$ s79 |
| s42 = s6 $\oplus$ s22  | s67 = s32 $\oplus$ s24 | s92 = s57 $\oplus$ s15  | s117 = s96 $\oplus$ s81 |
| s43 = s15 $\oplus$ s23 | s68 = s36 $\oplus$ s24 | s93 = s34 $\oplus$ s58  | s118 = s87 $\oplus$ s62 |
| s44 = s1 $\oplus$ s24  | s69 = s44 $\oplus$ s37 | s94 = s44 $\oplus$ s58  | s119 = s89 $\oplus$ s34 |
| s45 = s1 $\oplus$ s25  | s70 = s32 $\oplus$ s46 | s95 = s48 $\oplus$ s58  | s120 = s92 $\oplus$ s56 |
| s46 = s9 $\oplus$ s25  | s71 = s46 $\oplus$ s10 | s96 = s53 $\oplus$ s58  | s121 = s90 $\oplus$ s36 |
| s47 = s2 $\oplus$ s26  | s72 = s46 $\oplus$ s16 | s97 = s68 $\oplus$ s35  | s122 = s69 $\oplus$ s90 |
| s48 = s11 $\oplus$ s26 | s73 = s45 $\oplus$ s17 | s98 = s70 $\oplus$ s60  | s123 = s74 $\oplus$ s2  |
| s49 = s2 $\oplus$ s27  | s74 = s45 $\oplus$ s38 | s99 = s74 $\oplus$ s77  | s124 = s76 $\oplus$ s91 |
| s50 = s3 $\oplus$ s27  | s75 = s47 $\oplus$ s37 | s100 = s78 $\oplus$ s61 | s125 = s82 $\oplus$ s91 |
| s51 = s4 $\oplus$ s27  | s76 = s47 $\oplus$ s39 | s101 = s63 $\oplus$ s83 | s126 = s85 $\oplus$ s33 |
| s52 = s4 $\oplus$ s28  | s77 = s46 $\oplus$ s26 | s102 = s87 $\oplus$ s85 | s127 = s86 $\oplus$ s59 |
| s53 = s12 $\oplus$ s28 | s78 = s49 $\oplus$ s35 | s103 = s59 $\oplus$ s56 | s128 = s88 $\oplus$ s64 |
| s54 = s13 $\oplus$ s29 | s79 = s50 $\oplus$ s18 | s104 = s93 $\oplus$ s15 |                         |
| s55 = s21 $\oplus$ s29 | s80 = s50 $\oplus$ s38 | s105 = s65 $\oplus$ s67 |                         |
| s56 = s22 $\oplus$ s30 | s81 = s51 $\oplus$ s19 | s106 = s65 $\oplus$ s73 |                         |

Output signals (output bit  $j$  is carried by the listed signal,  $j = 0, \dots, 31$ ):

[97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128]

## B Circuit mixcolumns\_92gates\_depth4 (92 gates, depth 4)

Signals  $s_0, \dots, s_{31}$  are the inputs; gate  $k$  produces  $s_{32+k}$ . Canonical gate-list SHA-256 (see Section 5):

e79bfccf5322fe8933242b47b53e94e73163615c70d5c45c9721e27ac86fd97d

|                        |                        |                        |                        |
|------------------------|------------------------|------------------------|------------------------|
| s32 = s10 $\oplus$ s26 | s44 = s9 $\oplus$ s25  | s56 = s15 $\oplus$ s31 | s68 = s35 $\oplus$ s56 |
| s33 = s12 $\oplus$ s20 | s45 = s18 $\oplus$ s26 | s57 = s43 $\oplus$ s44 | s69 = s3 $\oplus$ s38  |
| s34 = s14 $\oplus$ s30 | s46 = s0 $\oplus$ s17  | s58 = s34 $\oplus$ s53 | s70 = s51 $\oplus$ s54 |
| s35 = s6 $\oplus$ s14  | s47 = s1 $\oplus$ s9   | s59 = s5 $\oplus$ s54  | s71 = s19 $\oplus$ s48 |
| s36 = s11 $\oplus$ s28 | s48 = s23 $\oplus$ s31 | s60 = s0 $\oplus$ s37  | s72 = s24 $\oplus$ s38 |
| s37 = s8 $\oplus$ s16  | s49 = s3 $\oplus$ s27  | s61 = s38 $\oplus$ s52 | s73 = s39 $\oplus$ s41 |
| s38 = s7 $\oplus$ s31  | s50 = s4 $\oplus$ s20  | s62 = s6 $\oplus$ s40  | s74 = s41 $\oplus$ s55 |
| s39 = s17 $\oplus$ s25 | s51 = s12 $\oplus$ s27 | s63 = s32 $\oplus$ s56 | s75 = s34 $\oplus$ s38 |
| s40 = s13 $\oplus$ s21 | s52 = s22 $\oplus$ s30 | s64 = s36 $\oplus$ s56 | s76 = s43 $\oplus$ s48 |
| s41 = s2 $\oplus$ s26  | s53 = s13 $\oplus$ s29 | s65 = s37 $\oplus$ s56 | s77 = s22 $\oplus$ s42 |
| s42 = s5 $\oplus$ s29  | s54 = s4 $\oplus$ s28  | s66 = s1 $\oplus$ s48  | s78 = s41 $\oplus$ s44 |
| s43 = s16 $\oplus$ s24 | s55 = s11 $\oplus$ s19 | s67 = s50 $\oplus$ s53 | s79 = s10 $\oplus$ s47 |



|                        |                         |                          |                          |
|------------------------|-------------------------|--------------------------|--------------------------|
| s80 = s45 $\oplus$ s49 | s92 = s58 $\oplus$ s62  | s104 = s63 $\oplus$ s49  | s116 = s47 $\oplus$ s109 |
| s81 = s21 $\oplus$ s33 | s93 = s60 $\oplus$ s76  | s105 = s64 $\oplus$ s50  | s117 = s52 $\oplus$ s107 |
| s82 = s67 $\oplus$ s81 | s94 = s35 $\oplus$ s77  | s106 = s64 $\oplus$ s69  | s118 = s96 $\oplus$ s104 |
| s83 = s57 $\oplus$ s66 | s95 = s15 $\oplus$ s61  | s107 = s23 $\oplus$ s75  | s119 = s33 $\oplus$ s106 |
| s84 = s58 $\oplus$ s77 | s96 = s69 $\oplus$ s74  | s108 = s71 $\oplus$ s80  | s120 = s71 $\oplus$ s105 |
| s85 = s42 $\oplus$ s81 | s97 = s65 $\oplus$ s72  | s109 = s46 $\oplus$ s72  | s121 = s103 $\oplus$ s51 |
| s86 = s59 $\oplus$ s67 | s98 = s38 $\oplus$ s60  | s110 = s102 $\oplus$ s66 | s122 = s97 $\oplus$ s100 |
| s87 = s59 $\oplus$ s40 | s99 = s70 $\oplus$ s71  | s111 = s18 $\oplus$ s101 | s123 = s90 $\oplus$ s75  |
| s88 = s10 $\oplus$ s73 | s100 = s57 $\oplus$ s46 | s112 = s89 $\oplus$ s78  |                          |
| s89 = s45 $\oplus$ s79 | s101 = s39 $\oplus$ s78 | s113 = s63 $\oplus$ s108 |                          |
| s90 = s23 $\oplus$ s68 | s102 = s39 $\oplus$ s65 | s114 = s108 $\oplus$ s55 |                          |
| s91 = s62 $\oplus$ s52 | s103 = s69 $\oplus$ s50 | s115 = s93 $\oplus$ s65  |                          |

Output signals (output bit  $j$  is carried by the listed signal,  $j = 0, \dots, 31$ ):

[97, 122, 89, 118, 119, 82, 84, 90, 115, 110, 111, 113, 120, 85, 91, 117, 93, 83, 88, 114, 99, 86, 92, 95, 98, 116, 112, 96, 121, 87, 94, 123]

## C Circuit mixcolumns\_88gates\_depth5\_fromscratch (88 gates, depth 5; from scratch, the frontier's depth-5 point)

*Status: found from scratch on this project's own lineage, with no imported material at any step; it is the depth-5 point of the frontier (Section 2). 88 is not a new gate count — it is Jean's [9], who has priority. What this circuit changes is that the (88, 5) point is now reached without depending on that circuit; it does not beat it. It is six gates below the published depth-5 94 [3] and two levels shallower than the published 88. Its depth 5 is forced, and its remove- $\leq 3$  shell is exhaustively empty (Section 4).*

Signals  $s_0, \dots, s_{31}$  are the inputs; gate  $k$  produces  $s_{32+k}$ . Canonical gate-list SHA-256 (see Section 5):

4fb3ba75a5e3208c8d853b0e26e289cdc979305a5c040ea38e3488db239df6ad

|                        |                        |                         |                           |
|------------------------|------------------------|-------------------------|---------------------------|
| s32 = s22 $\oplus$ s30 | s55 = s33 $\oplus$ s7  | s78 = s51 $\oplus$ s60  | s101 = s82 $\oplus$ s49   |
| s33 = s14 $\oplus$ s22 | s56 = s34 $\oplus$ s53 | s79 = s53 $\oplus$ s14  | s102 = s83 $\oplus$ s36   |
| s34 = s13 $\oplus$ s21 | s57 = s35 $\oplus$ s40 | s80 = s54 $\oplus$ s69  | s103 = s83 $\oplus$ s84   |
| s35 = s15 $\oplus$ s23 | s58 = s35 $\oplus$ s50 | s81 = s54 $\oplus$ s56  | s104 = s89 $\oplus$ s43   |
| s36 = s13 $\oplus$ s29 | s59 = s37 $\oplus$ s20 | s82 = s58 $\oplus$ s25  | s105 = s89 $\oplus$ s37   |
| s37 = s12 $\oplus$ s28 | s60 = s38 $\oplus$ s45 | s83 = s59 $\oplus$ s4   | s106 = s92 $\oplus$ s70   |
| s38 = s23 $\oplus$ s31 | s61 = s38 $\oplus$ s55 | s84 = s60 $\oplus$ s20  | s107 = s93 $\oplus$ s62   |
| s39 = s1 $\oplus$ s9   | s62 = s38 $\oplus$ s58 | s85 = s62 $\oplus$ s63  | s108 = s94 $\oplus$ s59   |
| s40 = s3 $\oplus$ s19  | s63 = s39 $\oplus$ s47 | s86 = s62 $\oplus$ s75  | s109 = s95 $\oplus$ s72   |
| s41 = s4 $\oplus$ s12  | s64 = s39 $\oplus$ s2  | s87 = s64 $\oplus$ s68  | s110 = s95 $\oplus$ s17   |
| s42 = s9 $\oplus$ s25  | s65 = s40 $\oplus$ s11 | s88 = s65 $\oplus$ s78  | s111 = s97 $\oplus$ s68   |
| s43 = s5 $\oplus$ s29  | s66 = s41 $\oplus$ s57 | s89 = s67 $\oplus$ s34  | s112 = s99 $\oplus$ s52   |
| s44 = s10 $\oplus$ s18 | s67 = s41 $\oplus$ s5  | s90 = s69 $\oplus$ s79  | s113 = s99 $\oplus$ s55   |
| s45 = s19 $\oplus$ s27 | s68 = s42 $\oplus$ s44 | s91 = s70 $\oplus$ s27  | s114 = s102 $\oplus$ s67  |
| s46 = s2 $\oplus$ s10  | s69 = s43 $\oplus$ s30 | s92 = s71 $\oplus$ s74  | s115 = s102 $\oplus$ s105 |
| s47 = s16 $\oplus$ s24 | s70 = s44 $\oplus$ s57 | s93 = s72 $\oplus$ s35  | s116 = s106 $\oplus$ s78  |
| s48 = s7 $\oplus$ s15  | s71 = s45 $\oplus$ s46 | s94 = s74 $\oplus$ s3   | s117 = s108 $\oplus$ s76  |
| s49 = s1 $\oplus$ s17  | s72 = s47 $\oplus$ s0  | s95 = s75 $\oplus$ s8   | s118 = s110 $\oplus$ s85  |
| s50 = s8 $\oplus$ s16  | s73 = s48 $\oplus$ s52 | s96 = s76 $\oplus$ s84  | s119 = s110 $\oplus$ s42  |
| s51 = s18 $\oplus$ s26 | s74 = s48 $\oplus$ s11 | s97 = s77 $\oplus$ s49  |                           |
| s52 = s32 $\oplus$ s31 | s75 = s48 $\oplus$ s0  | s98 = s77 $\oplus$ s46  |                           |
| s53 = s32 $\oplus$ s6  | s76 = s48 $\oplus$ s66 | s99 = s79 $\oplus$ s35  |                           |
| s54 = s33 $\oplus$ s36 | s77 = s51 $\oplus$ s64 | s100 = s82 $\oplus$ s85 |                           |

Output signals (output bit  $j$  is carried by the listed signal,  $j = 0, \dots, 31$ ):

[109, 119, 98, 92, 108, 104, 80, 112, 93, 101, 97, 91, 117, 115, 56, 61, 107, 100, 111, 88, 103, 114, 81, 73, 86, 118, 87, 116, 96, 105, 90, 113]

## D Circuit mixcolumns\_89gates\_depth5 (89 gates, depth 5)

Signals  $s_0, \dots, s_{31}$  are the inputs; gate  $k$  produces  $s_{32+k}$ . Canonical gate-list SHA-256 (see Section 5):

c3b972af282d25e8313a510f5cf840472401c6268bd28cda889593fc3379f064

|                        |                        |                         |                           |
|------------------------|------------------------|-------------------------|---------------------------|
| s32 = s9 $\oplus$ s17  | s55 = s21 $\oplus$ s39 | s78 = s55 $\oplus$ s66  | s101 = s60 $\oplus$ s50   |
| s33 = s19 $\oplus$ s27 | s56 = s5 $\oplus$ s46  | s79 = s37 $\oplus$ s57  | s102 = s39 $\oplus$ s101  |
| s34 = s1 $\oplus$ s25  | s57 = s14 $\oplus$ s42 | s80 = s35 $\oplus$ s56  | s103 = s34 $\oplus$ s87   |
| s35 = s21 $\oplus$ s29 | s58 = s15 $\oplus$ s37 | s81 = s56 $\oplus$ s66  | s104 = s59 $\oplus$ s89   |
| s36 = s12 $\oplus$ s28 | s59 = s11 $\oplus$ s33 | s82 = s55 $\oplus$ s42  | s105 = s76 $\oplus$ s70   |
| s37 = s22 $\oplus$ s30 | s60 = s11 $\oplus$ s43 | s83 = s70 $\oplus$ s72  | s106 = s92 $\oplus$ s65   |
| s38 = s3 $\oplus$ s27  | s61 = s0 $\oplus$ s52  | s84 = s43 $\oplus$ s71  | s107 = s96 $\oplus$ s98   |
| s39 = s4 $\oplus$ s28  | s62 = s33 $\oplus$ s45 | s85 = s54 $\oplus$ s67  | s108 = s36 $\oplus$ s93   |
| s40 = s9 $\oplus$ s25  | s63 = s38 $\oplus$ s52 | s86 = s54 $\oplus$ s49  | s109 = s95 $\oplus$ s48   |
| s41 = s13 $\oplus$ s29 | s64 = s43 $\oplus$ s51 | s87 = s17 $\oplus$ s64  | s110 = s1 $\oplus$ s92    |
| s42 = s5 $\oplus$ s13  | s65 = s8 $\oplus$ s45  | s88 = s57 $\oplus$ s67  | s111 = s99 $\oplus$ s46   |
| s43 = s15 $\oplus$ s23 | s66 = s36 $\oplus$ s41 | s89 = s60 $\oplus$ s47  | s112 = s98 $\oplus$ s75   |
| s44 = s6 $\oplus$ s22  | s67 = s41 $\oplus$ s44 | s90 = s62 $\oplus$ s53  | s113 = s89 $\oplus$ s90   |
| s45 = s7 $\oplus$ s23  | s68 = s34 $\oplus$ s53 | s91 = s71 $\oplus$ s74  | s114 = s108 $\oplus$ s99  |
| s46 = s12 $\oplus$ s20 | s69 = s24 $\oplus$ s45 | s92 = s24 $\oplus$ s61  | s115 = s110 $\oplus$ s112 |
| s47 = s10 $\oplus$ s18 | s70 = s40 $\oplus$ s47 | s93 = s38 $\oplus$ s62  | s116 = s3 $\oplus$ s104   |
| s48 = s2 $\oplus$ s26  | s71 = s31 $\oplus$ s49 | s94 = s58 $\oplus$ s52  | s117 = s90 $\oplus$ s109  |
| s49 = s6 $\oplus$ s14  | s72 = s26 $\oplus$ s34 | s95 = s59 $\oplus$ s63  | s118 = s108 $\oplus$ s101 |
| s50 = s19 $\oplus$ s20 | s73 = s32 $\oplus$ s48 | s96 = s64 $\oplus$ s69  | s119 = s87 $\oplus$ s112  |
| s51 = s8 $\oplus$ s16  | s74 = s44 $\oplus$ s45 | s97 = s61 $\oplus$ s51  | s120 = s32 $\oplus$ s110  |
| s52 = s7 $\oplus$ s31  | s75 = s40 $\oplus$ s51 | s98 = s0 $\oplus$ s65   |                           |
| s53 = s2 $\oplus$ s18  | s76 = s18 $\oplus$ s73 | s99 = s4 $\oplus$ s63   |                           |
| s54 = s30 $\oplus$ s35 | s77 = s10 $\oplus$ s68 | s100 = s58 $\oplus$ s74 |                           |

Output signals (output bit  $j$  is carried by the listed signal,  $j = 0, \dots, 31$ ):

[96, 119, 83, 113, 118, 78, 79, 84, 107, 103, 76, 116, 102, 80, 85, 91, 106, 115, 105, 117, 114, 81, 86, 94, 97, 120, 77, 109, 111, 82, 88, 100]

## E Circuit mixcolumns\_88gates\_depth5 (88 gates, depth 5; derived, superseded at its point)

*Status: derived from published work. Its seed chain passes through Jean's published 88-gate circuit [9] (Section 2), so this is not an independent construction and Jean is credited for the material it descends from. It is six gates below the published depth-5 94 [3], and it shares 75 of its 88 masks with Appendix D (Jaccard 0.735), the depth-5 89's own basin reached at 88 gates. 88 is Jean's count, not a new one. It is superseded at its own point by Appendix C and retained.*

Signals  $s_0, \dots, s_{31}$  are the inputs; gate  $k$  produces  $s_{32+k}$ . Canonical gate-list SHA-256 (see Section 5):

91d45c57d638dc5832a23eaf7721ba80d07bc0566769c47b56bd5566ae4eabff

|                        |                        |                        |                        |
|------------------------|------------------------|------------------------|------------------------|
| s32 = s13 $\oplus$ s29 | s46 = s1 $\oplus$ s25  | s60 = s39 $\oplus$ s4  | s74 = s48 $\oplus$ s31 |
| s33 = s21 $\oplus$ s29 | s47 = s6 $\oplus$ s22  | s61 = s40 $\oplus$ s50 | s75 = s50 $\oplus$ s8  |
| s34 = s22 $\oplus$ s30 | s48 = s6 $\oplus$ s14  | s62 = s40 $\oplus$ s51 | s76 = s50 $\oplus$ s58 |
| s35 = s15 $\oplus$ s23 | s49 = s11 $\oplus$ s19 | s63 = s41 $\oplus$ s21 | s77 = s51 $\oplus$ s0  |
| s36 = s9 $\oplus$ s17  | s50 = s7 $\oplus$ s23  | s64 = s42 $\oplus$ s45 | s78 = s51 $\oplus$ s56 |
| s37 = s12 $\oplus$ s20 | s51 = s7 $\oplus$ s31  | s65 = s42 $\oplus$ s52 | s79 = s51 $\oplus$ s60 |
| s38 = s12 $\oplus$ s28 | s52 = s8 $\oplus$ s16  | s66 = s43 $\oplus$ s26 | s80 = s52 $\oplus$ s77 |
| s39 = s3 $\oplus$ s27  | s53 = s32 $\oplus$ s38 | s67 = s43 $\oplus$ s61 | s81 = s53 $\oplus$ s63 |
| s40 = s3 $\oplus$ s19  | s54 = s32 $\oplus$ s47 | s68 = s44 $\oplus$ s14 | s82 = s53 $\oplus$ s59 |
| s41 = s4 $\oplus$ s28  | s55 = s33 $\oplus$ s30 | s69 = s44 $\oplus$ s63 | s83 = s54 $\oplus$ s55 |
| s42 = s9 $\oplus$ s25  | s56 = s34 $\oplus$ s15 | s70 = s45 $\oplus$ s11 | s84 = s54 $\oplus$ s68 |
| s43 = s2 $\oplus$ s18  | s57 = s35 $\oplus$ s49 | s71 = s46 $\oplus$ s26 | s85 = s56 $\oplus$ s72 |
| s44 = s5 $\oplus$ s13  | s58 = s35 $\oplus$ s52 | s72 = s47 $\oplus$ s50 | s86 = s57 $\oplus$ s20 |
| s45 = s10 $\oplus$ s18 | s59 = s37 $\oplus$ s5  | s73 = s48 $\oplus$ s55 | s87 = s58 $\oplus$ s17 |

|                        |                         |                           |                           |
|------------------------|-------------------------|---------------------------|---------------------------|
| s88 = s59 $\oplus$ s33 | s97 = s74 $\oplus$ s35  | s106 = s91 $\oplus$ s64   | s115 = s107 $\oplus$ s70  |
| s89 = s61 $\oplus$ s38 | s98 = s75 $\oplus$ s0   | s107 = s92 $\oplus$ s62   | s116 = s109 $\oplus$ s35  |
| s90 = s64 $\oplus$ s71 | s99 = s76 $\oplus$ s24  | s108 = s94 $\oplus$ s57   | s117 = s111 $\oplus$ s112 |
| s91 = s66 $\oplus$ s36 | s100 = s77 $\oplus$ s24 | s109 = s94 $\oplus$ s67   | s118 = s111 $\oplus$ s87  |
| s92 = s66 $\oplus$ s10 | s101 = s79 $\oplus$ s89 | s110 = s98 $\oplus$ s99   | s119 = s112 $\oplus$ s36  |
| s93 = s68 $\oplus$ s34 | s102 = s79 $\oplus$ s37 | s111 = s98 $\oplus$ s65   |                           |
| s94 = s70 $\oplus$ s39 | s103 = s86 $\oplus$ s41 | s112 = s100 $\oplus$ s1   |                           |
| s95 = s71 $\oplus$ s92 | s104 = s86 $\oplus$ s89 | s113 = s100 $\oplus$ s75  |                           |
| s96 = s72 $\oplus$ s74 | s105 = s87 $\oplus$ s46 | s114 = s107 $\oplus$ s109 |                           |

Output signals (output bit  $j$  is carried by the listed signal,  $j = 0, \dots, 31$ ):

[99, 118, 90, 116, 104, 81, 93, 97, 110, 105, 91, 108, 103, 88, 83, 96, 113, 117, 106, 114, 101, 82, 73, 78, 80, 119, 95, 115, 102, 69, 84, 85]

## F Circuit mixcolumns\_88gates\_depth6 (88 gates, depth 6; improves the frontier at depth 6)

*Status: found from scratch on this project's own lineage, with no imported circuit at any step. 88 is not a new gate count — it is Jean's [9], who has priority. What this circuit improves is the depth at that count: one level shallower than Jean's 88, and four gates below the published depth-6 92 [4]. Its remove- $\leq 3$  shell is exhaustively empty (Section 4). It is dominated within this note by Appendix C, a different family (Jaccard 0.323), and remains the first 88 found here from scratch.*

Signals  $s_0, \dots, s_{31}$  are the inputs; gate  $k$  produces  $s_{32+k}$ . Canonical gate-list SHA-256 (see Section 5):

462a149979fe0ab39173e6aa70c1728fe55cb24b380ea0b2241ffaa6301a8c4c

|                        |                        |                         |                           |
|------------------------|------------------------|-------------------------|---------------------------|
| s32 = s14 $\oplus$ s22 | s55 = s40 $\oplus$ s12 | s78 = s57 $\oplus$ s69  | s101 = s80 $\oplus$ s86   |
| s33 = s13 $\oplus$ s29 | s56 = s41 $\oplus$ s51 | s79 = s58 $\oplus$ s2   | s102 = s81 $\oplus$ s8    |
| s34 = s21 $\oplus$ s29 | s57 = s41 $\oplus$ s6  | s80 = s58 $\oplus$ s12  | s103 = s84 $\oplus$ s38   |
| s35 = s14 $\oplus$ s30 | s58 = s42 $\oplus$ s43 | s81 = s59 $\oplus$ s37  | s104 = s84 $\oplus$ s90   |
| s36 = s20 $\oplus$ s28 | s59 = s42 $\oplus$ s46 | s82 = s59 $\oplus$ s61  | s105 = s85 $\oplus$ s91   |
| s37 = s0 $\oplus$ s24  | s60 = s43 $\oplus$ s3  | s83 = s60 $\oplus$ s36  | s106 = s85 $\oplus$ s27   |
| s38 = s17 $\oplus$ s25 | s61 = s44 $\oplus$ s24 | s84 = s61 $\oplus$ s16  | s107 = s87 $\oplus$ s55   |
| s39 = s2 $\oplus$ s26  | s62 = s44 $\oplus$ s10 | s85 = s62 $\oplus$ s79  | s108 = s88 $\oplus$ s92   |
| s40 = s4 $\oplus$ s20  | s63 = s44 $\oplus$ s11 | s86 = s63 $\oplus$ s83  | s109 = s92 $\oplus$ s49   |
| s41 = s5 $\oplus$ s13  | s64 = s45 $\oplus$ s27 | s87 = s64 $\oplus$ s3   | s110 = s94 $\oplus$ s96   |
| s42 = s15 $\oplus$ s23 | s65 = s45 $\oplus$ s49 | s88 = s65 $\oplus$ s35  | s111 = s94 $\oplus$ s54   |
| s43 = s11 $\oplus$ s19 | s66 = s45 $\oplus$ s46 | s89 = s65 $\oplus$ s42  | s112 = s96 $\oplus$ s52   |
| s44 = s7 $\oplus$ s23  | s67 = s47 $\oplus$ s63 | s90 = s66 $\oplus$ s0   | s113 = s98 $\oplus$ s51   |
| s45 = s7 $\oplus$ s31  | s68 = s48 $\oplus$ s57 | s91 = s67 $\oplus$ s87  | s114 = s103 $\oplus$ s111 |
| s46 = s8 $\oplus$ s16  | s69 = s50 $\oplus$ s22 | s92 = s69 $\oplus$ s44  | s115 = s103 $\oplus$ s9   |
| s47 = s18 $\oplus$ s26 | s70 = s50 $\oplus$ s34 | s93 = s70 $\oplus$ s48  | s116 = s105 $\oplus$ s54  |
| s48 = s32 $\oplus$ s33 | s71 = s52 $\oplus$ s59 | s94 = s72 $\oplus$ s1   | s117 = s107 $\oplus$ s86  |
| s49 = s32 $\oplus$ s15 | s72 = s53 $\oplus$ s9  | s95 = s72 $\oplus$ s18  | s118 = s111 $\oplus$ s37  |
| s50 = s35 $\oplus$ s6  | s73 = s53 $\oplus$ s10 | s96 = s73 $\oplus$ s47  | s119 = s115 $\oplus$ s81  |
| s51 = s36 $\oplus$ s21 | s74 = s54 $\oplus$ s60 | s97 = s73 $\oplus$ s25  |                           |
| s52 = s38 $\oplus$ s1  | s75 = s55 $\oplus$ s28 | s98 = s75 $\oplus$ s33  |                           |
| s53 = s39 $\oplus$ s17 | s76 = s56 $\oplus$ s34 | s99 = s75 $\oplus$ s80  |                           |
| s54 = s39 $\oplus$ s45 | s77 = s56 $\oplus$ s40 | s100 = s77 $\oplus$ s98 |                           |

Output signals (output bit  $j$  is carried by the listed signal,  $j = 0, \dots, 31$ ):

[82, 119, 110, 106, 101, 113, 78, 108, 102, 71, 95, 116, 99, 100, 93, 89, 104, 114, 97, 91, 117, 76, 70, 88, 90, 118, 112, 74, 107, 77, 68, 109]

## G Circuit mixcolumns\_88gates\_depth7 (88 gates, depth 7; ties the published floor)

*Status: the same (depth, count) point as Jean [9], reached independently on this project's own lineage; 61 of 88 internal masks are shared. It ties that point and does not beat it, and Jean*

has priority. It is dominated by Appendices C and F, and kept because independent replication of a published point is worth archiving.

Signals  $s_0, \dots, s_{31}$  are the inputs; gate  $k$  produces  $s_{32+k}$ . Canonical gate-list SHA-256 (see Section 5):

76bac50ab0e4a7dd54578e77f57f6d08e96a8803aff72dba2c3f5fadaaa06ee1

|                                 |                                 |                                  |                                    |
|---------------------------------|---------------------------------|----------------------------------|------------------------------------|
| $s_{32} = s_{13} \oplus s_{29}$ | $s_{55} = s_{34} \oplus s_{15}$ | $s_{78} = s_{52} \oplus s_{64}$  | $s_{101} = s_{87} \oplus s_{37}$   |
| $s_{33} = s_{21} \oplus s_{29}$ | $s_{56} = s_{35} \oplus s_{20}$ | $s_{79} = s_{53} \oplus s_{63}$  | $s_{102} = s_{90} \oplus s_{61}$   |
| $s_{34} = s_{22} \oplus s_{30}$ | $s_{57} = s_{35} \oplus s_8$    | $s_{80} = s_{53} \oplus s_{59}$  | $s_{103} = s_{90} \oplus s_{40}$   |
| $s_{35} = s_{15} \oplus s_{23}$ | $s_{58} = s_{36} \oplus s_2$    | $s_{81} = s_{55} \oplus s_{68}$  | $s_{104} = s_{94} \oplus s_{38}$   |
| $s_{36} = s_{17} \oplus s_{25}$ | $s_{59} = s_{37} \oplus s_5$    | $s_{82} = s_{57} \oplus s_{16}$  | $s_{105} = s_{94} \oplus s_{40}$   |
| $s_{37} = s_{12} \oplus s_{20}$ | $s_{60} = s_{39} \oplus s_1$    | $s_{83} = s_{58} \oplus s_{61}$  | $s_{106} = s_{95} \oplus s_{60}$   |
| $s_{38} = s_{12} \oplus s_{28}$ | $s_{61} = s_{39} \oplus s_{51}$ | $s_{84} = s_{59} \oplus s_{33}$  | $s_{107} = s_{96} \oplus s_{44}$   |
| $s_{39} = s_9 \oplus s_{25}$    | $s_{62} = s_{41} \oplus s_{50}$ | $s_{85} = s_{60} \oplus s_{17}$  | $s_{108} = s_{96} \oplus s_{36}$   |
| $s_{40} = s_2 \oplus s_{18}$    | $s_{63} = s_{42} \oplus s_{21}$ | $s_{86} = s_{62} \oplus s_{76}$  | $s_{109} = s_{97} \oplus s_{108}$  |
| $s_{41} = s_3 \oplus s_{27}$    | $s_{64} = s_{43} \oplus s_{14}$ | $s_{87} = s_{62} \oplus s_4$     | $s_{110} = s_{98} \oplus s_{49}$   |
| $s_{42} = s_4 \oplus s_{28}$    | $s_{65} = s_{43} \oplus s_{63}$ | $s_{88} = s_{64} \oplus s_{34}$  | $s_{111} = s_{100} \oplus s_{105}$ |
| $s_{43} = s_5 \oplus s_{13}$    | $s_{66} = s_{44} \oplus s_{45}$ | $s_{89} = s_{66} \oplus s_{86}$  | $s_{112} = s_{104} \oplus s_{87}$  |
| $s_{44} = s_{15} \oplus s_{31}$ | $s_{67} = s_{45} \oplus s_{58}$ | $s_{90} = s_{67} \oplus s_{85}$  | $s_{113} = s_{104} \oplus s_{71}$  |
| $s_{45} = s_{10} \oplus s_{26}$ | $s_{68} = s_{46} \oplus s_{49}$ | $s_{91} = s_{68} \oplus s_{70}$  | $s_{114} = s_{106} \oplus s_{108}$ |
| $s_{46} = s_6 \oplus s_{22}$    | $s_{69} = s_{47} \oplus s_{54}$ | $s_{92} = s_{70} \oplus s_{35}$  | $s_{115} = s_{107} \oplus s_{74}$  |
| $s_{47} = s_6 \oplus s_{14}$    | $s_{70} = s_{47} \oplus s_{31}$ | $s_{93} = s_{71} \oplus s_{42}$  | $s_{116} = s_{107} \oplus s_{110}$ |
| $s_{48} = s_{11} \oplus s_{19}$ | $s_{71} = s_{48} \oplus s_{56}$ | $s_{94} = s_{72} \oplus s_3$     | $s_{117} = s_{107} \oplus s_{57}$  |
| $s_{49} = s_7 \oplus s_{23}$    | $s_{72} = s_{49} \oplus s_{19}$ | $s_{95} = s_{73} \oplus s_{74}$  | $s_{118} = s_{111} \oplus s_{41}$  |
| $s_{50} = s_7 \oplus s_{31}$    | $s_{73} = s_{49} \oplus s_{16}$ | $s_{96} = s_{73} \oplus s_0$     | $s_{119} = s_{111} \oplus s_{66}$  |
| $s_{51} = s_{18} \oplus s_{26}$ | $s_{74} = s_{50} \oplus s_{24}$ | $s_{97} = s_{82} \oplus s_9$     |                                    |
| $s_{52} = s_{32} \oplus s_{46}$ | $s_{75} = s_{50} \oplus s_{55}$ | $s_{98} = s_{82} \oplus s_{24}$  |                                    |
| $s_{53} = s_{32} \oplus s_{38}$ | $s_{76} = s_{51} \oplus s_{72}$ | $s_{99} = s_{85} \oplus s_{97}$  |                                    |
| $s_{54} = s_{33} \oplus s_{30}$ | $s_{77} = s_{52} \oplus s_{54}$ | $s_{100} = s_{86} \oplus s_{48}$ |                                    |

Output signals (output bit  $j$  is carried by the listed signal,  $j = 0, \dots, 31$ ):

[110, 109, 103, 119, 113, 79, 88, 92, 115, 99, 83, 89, 93, 84, 77, 91, 116, 106, 67, 100, 112, 80, 69, 75, 117, 114, 102, 118, 101, 65, 78, 81]

## H Circuit mixcolumns\_88gates\_depth8 (88 gates, depth 8; derived, non-frontier)

*Status: derived from published work, and not a frontier point. Its seed chain passes through Jean's published 88-gate circuit [9] (Section 2), and it is dominated by every other 88 here (same count, greater depth). It is listed as a verified distinct construction, not as a claim.*

Signals  $s_0, \dots, s_{31}$  are the inputs; gate  $k$  produces  $s_{32+k}$ . Canonical gate-list SHA-256 (see Section 5):

3a23e2f2b2f3835cab74ba23800088bb03ab4ee3f076f2b9a826c78cd84f8ef1

|                                 |                                 |                                 |                                 |
|---------------------------------|---------------------------------|---------------------------------|---------------------------------|
| $s_{32} = s_{13} \oplus s_{29}$ | $s_{49} = s_7 \oplus s_{23}$    | $s_{66} = s_{48} \oplus s_{31}$ | $s_{83} = s_{64} \oplus s_{66}$ |
| $s_{33} = s_{21} \oplus s_{29}$ | $s_{50} = s_7 \oplus s_{31}$    | $s_{67} = s_{49} \oplus s_3$    | $s_{84} = s_{66} \oplus s_{35}$ |
| $s_{34} = s_{22} \oplus s_{30}$ | $s_{51} = s_{32} \oplus s_{37}$ | $s_{68} = s_{50} \oplus s_{24}$ | $s_{85} = s_{67} \oplus s_{19}$ |
| $s_{35} = s_{15} \oplus s_{23}$ | $s_{52} = s_{32} \oplus s_{47}$ | $s_{69} = s_{50} \oplus s_{54}$ | $s_{86} = s_{67} \oplus s_{11}$ |
| $s_{36} = s_9 \oplus s_{17}$    | $s_{53} = s_{33} \oplus s_{30}$ | $s_{70} = s_{51} \oplus s_{58}$ | $s_{87} = s_{68} \oplus s_0$    |
| $s_{37} = s_{12} \oplus s_{28}$ | $s_{54} = s_{34} \oplus s_{15}$ | $s_{71} = s_{52} \oplus s_{53}$ | $s_{88} = s_{74} \oplus s_{17}$ |
| $s_{38} = s_2 \oplus s_{18}$    | $s_{55} = s_{35} \oplus s_8$    | $s_{72} = s_{52} \oplus s_{60}$ | $s_{89} = s_{76} \oplus s_{44}$ |
| $s_{39} = s_3 \oplus s_{27}$    | $s_{56} = s_{38} \oplus s_{26}$ | $s_{73} = s_{54} \oplus s_{64}$ | $s_{90} = s_{77} \oplus s_{41}$ |
| $s_{40} = s_4 \oplus s_{28}$    | $s_{57} = s_{39} \oplus s_{50}$ | $s_{74} = s_{55} \oplus s_{16}$ | $s_{91} = s_{81} \oplus s_{33}$ |
| $s_{41} = s_4 \oplus s_{20}$    | $s_{58} = s_{40} \oplus s_{21}$ | $s_{75} = s_{56} \oplus s_{36}$ | $s_{92} = s_{81} \oplus s_{51}$ |
| $s_{42} = s_9 \oplus s_{25}$    | $s_{59} = s_{42} \oplus s_{45}$ | $s_{76} = s_{56} \oplus s_{10}$ | $s_{93} = s_{85} \oplus s_{89}$ |
| $s_{43} = s_5 \oplus s_{13}$    | $s_{60} = s_{43} \oplus s_{14}$ | $s_{77} = s_{57} \oplus s_{12}$ | $s_{94} = s_{85} \oplus s_{40}$ |
| $s_{44} = s_{15} \oplus s_{31}$ | $s_{61} = s_{43} \oplus s_{58}$ | $s_{78} = s_{59} \oplus s_{75}$ | $s_{95} = s_{86} \oplus s_{35}$ |
| $s_{45} = s_{10} \oplus s_{18}$ | $s_{62} = s_{45} \oplus s_{19}$ | $s_{79} = s_{59} \oplus s_{63}$ | $s_{96} = s_{87} \oplus s_{49}$ |
| $s_{46} = s_1 \oplus s_{25}$    | $s_{63} = s_{46} \oplus s_{26}$ | $s_{80} = s_{60} \oplus s_{34}$ | $s_{97} = s_{87} \oplus s_1$    |
| $s_{47} = s_6 \oplus s_{22}$    | $s_{64} = s_{47} \oplus s_{49}$ | $s_{81} = s_{61} \oplus s_{41}$ | $s_{98} = s_{88} \oplus s_{46}$ |
| $s_{48} = s_6 \oplus s_{14}$    | $s_{65} = s_{48} \oplus s_{53}$ | $s_{82} = s_{63} \oplus s_{76}$ | $s_{99} = s_{94} \oplus s_{77}$ |

|                                   |                                   |                                    |                                   |
|-----------------------------------|-----------------------------------|------------------------------------|-----------------------------------|
| $s_{100} = s_{95} \oplus s_{62}$  | $s_{106} = s_{100} \oplus s_{39}$ | $s_{112} = s_{106} \oplus s_{38}$  | $s_{118} = s_{114} \oplus s_{97}$ |
| $s_{101} = s_{95} \oplus s_{20}$  | $s_{107} = s_{101} \oplus s_{94}$ | $s_{113} = s_{109} \oplus s_{68}$  | $s_{119} = s_{114} \oplus s_{88}$ |
| $s_{102} = s_{96} \oplus s_{16}$  | $s_{108} = s_{101} \oplus s_{37}$ | $s_{114} = s_{110} \oplus s_{42}$  |                                   |
| $s_{103} = s_{96} \oplus s_8$     | $s_{109} = s_{102} \oplus s_{44}$ | $s_{115} = s_{112} \oplus s_{93}$  |                                   |
| $s_{104} = s_{97} \oplus s_{36}$  | $s_{110} = s_{102} \oplus s_{68}$ | $s_{116} = s_{113} \oplus s_{103}$ |                                   |
| $s_{105} = s_{100} \oplus s_{89}$ | $s_{111} = s_{106} \oplus s_{86}$ | $s_{117} = s_{113} \oplus s_{55}$  |                                   |

Output signals (output bit  $j$  is carried by the listed signal,  $j = 0, \dots, 31$ ):

[116, 119, 79, 112, 108, 70, 80, 84, 109, 98, 75, 111, 107, 92, 71, 83, 103, 118, 78, 115, 99, 91, 65,  
69, 117, 104, 82, 105, 90, 61, 72, 73]